maternova™
Solutions for mothers & newborns
Soluciones para madres y recién nacidos

The MaternaWell Tray™

Making the 'E' in E-MOTIVE Eco-Friendly and Cost-Effective

Postpartum Hemorrhage is the leading cause of preventable maternal deaths worldwide. Objective assessment of blood loss is a very important step to improve outcomes. The 2023 E-MOTIVE study demonstrated that the early detection of and quantification of blood loss is essential, by demonstrating a 60 percent reduction in severe blood loss, surgery or mortality using the E-MOTIVE bundle. In this analysis we demonstrate that as compared to the calibrated plastic drape, a reusable blood loss measurement tool, the MaternaWell Tray™, is:

30x more cost-effective

150x more eco-friendly

Over 30x less expensive

The MaternaWell Tray™ is a patented, calibrated, comfortable tray that is tucked under a woman's buttocks post delivery to catch and measure blood loss.

The tray was invented by and purpose-built by a renowned obstetrician in response to the need to create a sustainable, reusable solution to measuring blood loss.

The cost of moving consumables across Africa is at least 30 to 50 percent of the cost of the product. Using Uganda as an example, we can see that though the initial cost per drape is a fraction of the tray, the need to resupply the drape for every birth starts to drive a cost differential. An average Ugandan district will need 10,000 calibrated drapes per year whereas the tray, because it is re-usable and autoclavable can be supplied as 81 units in total. In the first year, a district will spend U$16,500 on drapes versus U$2,527 on trays. This is a cost of U$1.65 per birth for the drape versus U$0.25 per birth for the tray. After 5 years, the savings is more apparent– U$1.65 per birth for the drape versus $.08 for the tray.

Extrapolated to the country, comprised of 146 districts, expenditure would be U$12M on the drapes versus U$368,971 on the trays over five years.

Table 1: 32x Higher Cost of Drape: Cost comparison of Calibrated V-Drape versus MaternaWell Tray™ in a landlocked African Country (U$D)[1]

Landlocked Country in Africa (e.g. Uganda, Chad)	Calibrated V-drapes	MaternaWell Tray
Cost per Unit	$0.70	$15.00
Shipping/Logistics[2] per unit	$0.80	$16.00
Storage per unit	$0.05	$0.10
Disinfection/sterilization	$0.00	$0.10
Medical waste management	$0.10	$0.00
Subtotal per unit cost	$1.65	$31.20
Avg number of Products/ district/ year[3]	10,000	81
Total Cost per district per year	$16,500	$2,527
Cost per birth year 1	$1.65	$0.25
5 year total cost per district[4]	$82,500	$2,527
5 year cost for the country (assumes 146 districts)	$12,045,000	$368,971
Cost per birth: 5 years	**$1.65**	**$0.08**

Assumptions

[1] 146 Districts in Uganda. 34.5 births per 1000 people. Population of 47.5 million =1.6 million births per year. 27.4 births per day per district
[2] Assumes low volume of trays shipped. MWT shipping costs to region would be significantly reduced at scale
[3] Assumes 27 births per day. Calculation is 2 spare trays for every tray in use
[4] Assumes MaternaWell trays last for 5 years under repeated use (per experience in South Africa)

Medical waste management / environmental impact fees not factored in here

Over 150x lower CO_2 Emissions with the Tray

Once the tray is in place it can be used for years, meaning no risk of stock-outs. As a social enterprise engaged in global trade of medical devices and consumables, we are acutely aware of the additional costs associated with resupply of a consumable. In calculating the estimated cost of transporting the calibrated drape using Uganda as a representative country we find that though the CO^2 emissions of the raw material and manufacture of tray versus drape are comparable (2.3 v. 2.7 kg CO_2e/kg product), the shipping and resupply of the product drives a radically different story. Using Uganda as an example, the cost of shipping, resupplying, manufacturing enough drapes and disposing of/incinerating results in a massive difference– per district– 6170 kg CO_2e versus 173 kg CO_2e with the tray.

In the overall context of decarbonization, a re-usable solution exists and offers a more affordable alternative to the drape.

Blood loss volume estimated

Blood loss may be estimated from the calibrations in the wells as it happens, giving real time estimation.

Table 2: 161X Higher CO_2 Emissions: Rough Annual CO_2 Emission

Estimates Comparing Drape to MaternaWell Tray™ for Uganda (one district)

Measurement	Unit of Measure	Drape	MWT
Unit Mass	kg	0.075	0.4
Units per annum		10000	81
Mass per annum	kg	750	32.4
CO_2e raw material + manufacture	kg CO_2e/kg product	2.3	2.7
Transport			
Road transport Sasolburg - JHB (100km)	kg CO_2e/kg product	0.022	0.022
Air freight JHB - Kampala (3000km)	kg CO_2e/kg product	2.4	2.4
Road transport Kamapala to district (250km)	kg CO_2e/kg product	0.055	0.055
Disposal			
Disposal Mass per annum	kg	750	0
Landfill emissions of plastic[1]	kg CO_2e/kg product	1.5	1.5
Incineration emissions of plastic[2]	kg CO_2e/kg product	2.7	2.9
Recyling emissions of plastic	kg CO_2e/kg product	0.75	0.75
Emissions			
Annual Manufacturing emissions	kg CO_2e	1725	87
Annual transport emissions	kg CO_2e	1858	80
Annual disposal emissions	kg CO_2e	2588	0
Annual year recycling emissions	kg CO_2e	0	5
Total Emissions/ Ugandan District/ year 1[3]	**kg CO_2e**	**6170**	**173**
Total Emissions/ Ugandan District/ 5 years[4]	**kg CO_2e**	**30851.25**	**192**

Assumptions

[1] assume 50% landfill
[2] assume 50% incineration
[3] 36 times higher CO_2 emissions for drape
[4] 161 times higher CO_2 emissions for drape over 5 years

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